

Arboretum Eco-Quest

Solve outdoor ecology checkpoints

At a glance: Read the living collection and solve the checkpoints before time runs out. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: tree identification, ecosystems, observation, classification evidence (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- 4 sets of clue cards, one card per checkpoint (plus 1 spare set)
- 1 laminated dichotomous key per team, keyed to the route (plus 1 spare; build it in prep, currently MISSING from the buy list)
- 4 route maps with the checkpoints marked (plus 2 spares)
- 4 clipboards (plus 1 spare)
- 4 evidence token bags (plus 1 spare), with evidence tokens for each checkpoint (about 30 tokens; currently MISSING from the buy list)
- pencils, shared
- optional QR codes that open the Arboretum Explorer record at each checkpoint

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready.
- 02 Stage the scoring sheet and any reference values before testing begins.
- 03 No PPE: this is an outdoor observation walk with no chemicals, water, soil, or instruments. Set up sun and hydration support, run the weather and heat-index check, confirm the buddy system and route boundaries, and confirm the approved route with the arboretum host.
- 04 Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Read the living collection and solve the checkpoints before time runs out." Take a quick prediction by show of hands.

Phase 2 Predict and plan the route

0:05 - 0:12

Each team takes a quick prediction about which clue will be most reliable, then numbers the checkpoints on the route map in an order that avoids backtracking. No walking yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Walk and solve

0:18 - 0:44

Teams walk their route, use the clue cards and the dichotomous key to answer each checkpoint, record the justifying clue, and collect the evidence token. Circulate, ask which clue led to each answer, and resist naming the tree for them.

Phase 5 Record evidence and collect tokens

At each checkpoint, teams write the one specific clue that justifies their answer and take the evidence token.

Phase 6 Check answers

Before returning, each team confirms every answer cites a real observed clue, not a guess, and has a token for each solved checkpoint.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Collect the route packs, the keys, the clipboards, and the evidence tokens; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team backtracks and wastes time. **Fix:** Have them re-number the remaining checkpoints into a no-backtrack order; treat it as a route-planning coaching moment, since pace is scored.

Problem: A team is stuck at a checkpoint and cannot reach a name. **Fix:** Send them back to the first either-or choice on the key, have them point to the actual leaf or bark feature that decides it, and use the optional QR code only to confirm after they commit to an answer.

Problem: A team answers from memory instead of the clues. **Fix:** Require them to write the specific feature (leaf shape, bark, branching, or seed) for each answer; an answer with no cited clue scores zero on evidence quality.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to observation as evidence.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	100

CLEANUP AND SOURCES

Collect the route maps, clue cards, dichotomous keys, clipboards, and evidence tokens and return them to the bin, and collect score slips. Nothing is consumed and there are no surfaces to wipe.

SOURCES Ambler Arboretum self-guided tour (<https://ambler.temple.edu/arboretum/our-gardens/self-guided-tour>); Arboretum Explorer interactive collection map (<https://ambler.arboretumexplorer.org/>)

Checkpoint clue cards

Print ONE set per team (4) plus 1 spare set, then cut apart. Staff write which tagged tree or post each numbered checkpoint is; teams record their identification and the one clue that decided it. Laminating is optional; teams write on these, so plain cardstock is fine.

Checkpoint 1

token ○

Tree or marker at this post (staff fill):

Your ID for this tree: _____

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 2

token ○

Tree or marker at this post (staff fill):

Your ID for this tree: _____

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 3

token ○

Tree or marker at this post (staff fill):

Your ID for this tree: _____

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 4

token ○

Tree or marker at this post (staff fill):

Your ID for this tree: _____

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 5

token ○

Tree or marker at this post (staff fill):

Your ID for this tree: _____

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 6

token ○

Tree or marker at this post (staff fill):

Your ID for this tree: _____

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Tree dichotomous key

A general temperate-tree key. Staff: trim or extend it so every couplet ends at a tree that is actually on your route, then print 1 per team (4) plus 1 spare and laminate for reuse. Start at 1 and follow the choice that matches what you see.

STEP	WHAT TO LOOK FOR	THEN
1a	Leaves are needles or flat scales (evergreen)	go to 2
1b	Leaves are broad and flat	go to 4
2a	Needles in bundles or clusters	Pine
2b	Needles attached singly to the twig	go to 3
3a	Needles flat and soft, will not roll between your fingers	Fir or Hemlock
3b	Needles four-sided and sharp, roll between your fingers	Spruce
4a	Leaves grow in opposite pairs	go to 5
4b	Leaves grow alternately (staggered)	go to 6
5a	Leaf is one blade with palm-like lobes	Maple
5b	Leaf is divided into leaflets	Ash
6a	Leaf is divided into many leaflets	go to 7
6b	Leaf is a single blade	go to 8
7a	Small leaflets, twig may have thorns	Locust
7b	Large leaflets, no thorns	Hickory or Walnut
8a	Leaf margin has teeth	go to 9
8b	Leaf margin is smooth or lobed	go to 10
9a	Bark white or papery, peels in strips	Birch
9b	Bark smooth gray, buds long and pointed	Beech
10a	Leaf has rounded or pointed lobes	Oak
10b	Leaf is heart-shaped, margin smooth	Redbud or Katsura

Route map (staff template)

TEMPLATE — needs your input. This map is a placeholder: the checkpoints are numbered but not placed, because the route is specific to your approved arboretum path. Staff: draw the paths between the posts, move or renumber the checkpoints to match the ground, and write the tree or garden at each. Print 4 (plus 2 spares) once finalized. Base map: Ambler Arboretum self-guided tour and Arboretum Explorer.

START / FINISH

●

1

tree: _____

2

tree: _____

3

tree: _____

4

tree: _____

5

tree: _____

6

tree: _____

Evidence tokens (cut apart)

Cut apart. Each team collects one token per solved checkpoint. This sheet makes 30 tokens (4 teams + 1 spare set of 6). Print on cardstock; no lamination needed.

